

VitalSoC Block-Level Compliance Audit

Source of truth: VitalSoC Register Specification, Rev 0.1 (draft), 8 Oct 2026 (cited as "Spec §x"). **Scope:** static review of the supplied files only: `timer.v`, `gpio.v`, `fir_periph.v`, `fir_core.v`, `fir.h`, `i2c_wb_bridge.v`, `i2c_master_top.v`, `i2c_master_byte_ctrl.v`, `i2c_master_bit_ctrl.v`, `i2c_master_defines.v`, and the architecture diagram. **Not done:** no simulation, lint or synthesis (no project directory, testbenches, golden model or `coeffs.h` supplied). Nothing was modified, integrated or fixed. **Line numbers:** the sources were supplied inline without line numbering, so findings cite the construct or code region instead of a line number.

Why is Wishbone in the I2C path?

Spec §8 says the I2C master is "the OpenCores `i2c_master_top` behind a bus bridge". The OpenCores core has only one host interface: an 8-bit, 3-bit-address **Wishbone slave**. PicoRV32's native `mem_valid/mem_ready` bus is not Wishbone, so `i2c_wb_bridge.v` converts one into the other.

Wishbone is purely internal. The bridge exposes the same `sel/wstrb/addr/wdata/rdata/ready/irq` slave convention as the other peripherals, plus `scl_i/sda_i` and `scl_drive_low/sda_drive_low`. No Wishbone signals leave the bridge. Nothing is wrong here, and replacing it would mean rewriting the OpenCores core. Your diagram labels the block "I2C WB bridge", which is consistent.

A. Specification Compliance Matrix

Required Block	RTL Module/File	Required Features	Verification Result	Issues Found
Boot ROM 2 KB @ 0x0000_0000	not supplied	read-only, resident bootloader (§2.1, §3)	UNVERIFIED	No RTL supplied
App RAM 64 KB @ 0x1000_0000	not supplied	64 KB, spec says the bootloader build is 8 KB (§2.1, §13 #2)	UNVERIFIED	No RTL supplied
UART @ 0x2000_0000	not supplied	DATA/STATUS, 115200 baud, CLKS_PER_BIT=868 (§4)	UNVERIFIED	No RTL supplied. Spec §2.2 says the current RTL has a different UART
GPIO @ 0x3000_0000	<code>gpio.v</code>	LED RW[15:0], SW R[15:0], BTN R[4:0], 2-flop sync, no debounce, DEADBEEF on unmapped (§5)	PASS (static)	Concerns only: G-1, G-2
Timer @ 0x4000_0000	<code>timer.v</code>	CTRL, PERIOD, FLAG (RW1C), TICKS, level IRQ (§6)	PARTIAL	T-1 confirmed (set/clear priority reversed); T-2, T-3
FIR peripheral wrapper @ 0x5000_0000	<code>fir_periph.v</code>	6 registers, W1C DONE, level IRQ, DEADBEEF on unmapped (§7.2–7.8)	PARTIAL	F-1 confirmed (0x18/0x1C read 0, not 0xDEADBEEF); F-3
FIR core (4 ch × 64 taps, shared MAC)	<code>fir_core.v</code>	Q1.15, 40-bit acc, >>>15, sat16, ~67 cycles, reset keeps coef/delay/pointers (§7.1, §7.9)	PASS (static trace)	F-2 confirmed (<code>dout</code> has no reset); bit-exactness vs <code>sw_fir_step</code> is unverified
I2C bridge @ 0x6000_0000	<code>i2c_wb_bridge.v</code>	Offsets 0x00–0x10, bits 31:8 read 0, DEADBEEF on	PASS (static)	Concerns only: I-7, I-9

Required Block	RTL Module/File	Required Features	Verification Result	Issues Found
		unmapped, irq[5] (§8.1)		
I2C master core (OpenCores)	<code>i2c_master_top/byte_ctrl/bit_ctrl/defines.v</code>	Standard OpenCores register set, PRER reset 0xFFFF, CTR/CR/SR bit layout (§8.2-8.6)	PARTIAL	I-1 confirmed (reserved CTR bits are RW); I-3, I-4
Address decoder + IRQ collector	not supplied	New map (§2.1), irq[3..6] (§2.2)	UNVERIFIED	Integration-owned; spec §13 #1 says the old decoder must be moved

Out of count (STRETCH, no spec issued): MI modem @ 0x7000_0000 / irq[6], and the PCPI `fir.step` custom instruction (Spec §10).

Firmware deliverable (supplied): `fir.h` is **PARTIAL**; see FH-1 to FH-3.

B. Detailed Findings

B1. Timer (`timer.v`)

Matches the spec:

- Reset values: PERIOD=99,999, ENABLE=0, IRQ_EN=0, TICK=0, TICKS=0 (§6.2-6.5).
- Tick interval = PERIOD+1 cycles (`cnt` runs 0..PERIOD).
- With ENABLE=0 the counter is held at 0, TICK stays set, and TICKS is not cleared.
- `irq = pending & irq_en` is a level IRQ.
- TICKS is read-only; FLAG write-1-clear.
- Register read formats are correct.

Confirmed defects

- T-1 (confirmed): set-vs-clear priority is reversed.**
 - Spec §6.4: "If hardware sets TICK in the same cycle as a clear, the set wins."
 - In the tick branch, `pending <= 1` is suppressed by `if (!(xfer && wr && addr==2 && wstrb[0] && wdata[0]))`. The comment says "don't overwrite a software clear", so the clear wins. This is the opposite of the spec, and the tick is lost (TICKS still increments).
 - Correction:** remove the guard so `pending <= 1'b1` is unconditional in the tick branch. It already sits after the write block, so the last non-blocking assignment wins.

Potential concerns

- **T-2 (dead code):** `cnt <= 32'd0` in the PERIOD write case never takes effect when `en=1`. The counting block later in the same `always` assigns `cnt` on every path, and the last non-blocking assignment wins.
 - The resulting behaviour does match Spec §6.3, where a PERIOD below the current count ticks immediately, because the next `cnt >= period` compare uses the new value. But it is compliant by accident, and it contradicts the author's visible intent.
 - Remove or document the line. Do not "fix" it by making the clear effective, since that would break §6.3.
- **T-3 (decode/aliasing):** `addr` is `[1:0]`, so all four word offsets are mapped and every offset above 0x0C aliases onto them.
 - Spec §1 allows aliasing within the 4 KB window, but also says unmapped accesses return 0xDEADBEEF. See Ambiguity A-1.
 - The file does not document which `mem_addr` bits feed `addr`; the integrator must infer `mem_addr[3:2]`.
- **T-4:** `rdata` has no reset and updates every cycle regardless of `sel`. The integrator must mux read data by region select and must drop `sel` the cycle after `ready`.
- **Verification:** the spec marks Timer WRITTEN, not simulated, and no testbench was supplied.

B2. GPIO (`gpio.v`)

Matches the spec:

- LED RW [15:0] with byte-lane writes on `wstrb[0]/[1]` and reset 0.
- SW read [15:0] and BTN read [4:0], through 2-flop synchronizers.
- Upper bits read 0.
- Writes to SW/BTN are ignored.
- Offsets 0x0C-0x1C read 0xDEADBEEF.
- One-cycle ready. `addr` is `mem_addr[4:2]`, documented.
- No debounce, as the spec states.

Potential concerns (not defects)

- **G-1:**
 - Spec §5 and §11 say non-existent switches and buttons read 0 and LED bits above the board's count have no effect. The RTL is fixed at 16/16/5 and has no board masking.
 - The top level must tie `btn[4]` low on Boolean (4 buttons) and truncate `led` / `sw` for ZedBoard (8). Otherwise the "reads 0" rule fails at board level.
 - The synchronizer flops have no `ASYNC_REG` attribute, and they are reset to 0, which is acceptable.
- **G-2:** `addr` is 3 bits, so offsets 0x20+ alias onto 0x00-0x1C (see A-1).

B3. FIR wrapper (`fir_periph.v`)

Matches the spec:

- CTRL write always updates CHANNEL[9:8] and IRQ_EN; START is self-clearing, reads 0, and is ignored while BUSY.
- Accepting a start clears DONE; a rejected start does not.
- STATUS reads {DONE, BUSY} with write-1-clear on DONE.
- DATA_IN takes `wdata[15:0]` and reads 0.

- DATA_OUT is sign-extended and does not clear DONE.
- COEF_ADDR is {ch[7:6], tap[5:0]} and reads 0.
- COEF_DATA triggers the store one cycle later via `coef_we`.
- `irq = done_s & irq_en` is a level IRQ.
- Reset clears CTRL/STATUS/IRQ_EN/DIN/CADDR.

Confirmed defects

- **F-1:** offsets 0x18 and 0x1C fall into `default: ;` and read 0.
 - Spec §1 requires unmapped reads to return 0xDEADBEEF. GPIO and the I2C bridge do this correctly, so FIR is inconsistent with its siblings.
 - **Correction:** in the `default` branch set `rdata <= 32'hDEADBEEF`.

Potential concerns

- **F-3:** a write with any non-zero `wstrb` is treated as a full-word write. A partial strobe, such as a `sb` to byte 3, would overwrite CHANNEL and IRQ_EN from `wdata`. Spec §1 forbids sub-word access, but nothing defends against it.
- **F-5:**
 - Coefficient writes during BUSY are not blocked. This is a documented firmware rule (§7.10).
 - If a software DONE clear coincides with the core's one-cycle `done` pulse, the clear wins and the completion is lost. The window is only reachable by clearing DONE mid-step, as `fir_init`'s timeout path could.
- **F-6:** `addr[1:0]` is unused, and `addr` is a raw `mem_addr[4:0]` rather than a word index like the other peripherals. See Section D.

B4. FIR core (`fir_core.v`)

I traced the pipeline by hand, edge by edge:

- Start accepted at E0, MACs at E3..E66 (64 products), and `done` at E67. That is **67 cycles = TAPS+3**, matching Spec §7.1. The file's header comment says "TAPS+4", which is a documentation error only.
- Tap 0 reads `dly[head]`, the newest sample, which is written at E0 and visible to the E1 read. ✓ (§7.7)
- Arithmetic checks out:
 - Parameters: ACC_W=40, DATA_W=COEF_W=16, COEF_FRAC=15.
 - Signed product, arithmetic `>>>15`, saturation limits $\pm 32767/-32768$.
- Reset clears `busy`, `issuing`, `acc` and `k`, but not `coef`, `dly` or `head`. ✓ (§7.9)
- `ch_sel` / `din_r` are stable across the start pulse, so the core's input latching is safe.

Confirmed defect

- **F-2:** `dout` is not reset and has no initial value. Spec §7.6 gives DOUT a reset value of 0.
 - In simulation DOUT is X until the first step completes.
 - On hardware, a CPU reset without bitstream reload leaves the previous output visible.
 - **Correction:** reset `dout` (or clear it in the wrapper's read path).

Unverified: the spec claims SIM-VERIFIED and bit-exact, but no testbench, `fir_golden.py`, `dsp.h` or `coeffs.h` was supplied, so bit-exactness is not confirmed.

B5. FIR driver header (`fir.h`)

Register offsets, bit positions and the coefficient address encoding all match the spec.

- **FH-1 (conflict):**
 - The header cites "Rev 0.2" and "Section 8.2 / 8.9 / 8.10" for FIR. The PDF is Rev 0.1 and FIR is §7 (§7.2, 7.9, 7.10). §8 is I2C.
 - Either a Rev 0.2 exists that was not supplied, or the references are wrong.
- **FH-2:**
 - `fir_init` is `void` and its timeout silently falls through, which violates the §12 rule "every wait has a timeout and returns an error".
 - `fir_step` swallows the timeout and returns 0.
 - `fir_step_poll(..., timeout_cycles=0)` waits forever, which violates the "no infinite loops" rule.
- **FH-3:** `fir_init` is safe to call twice, as required.

B6. I2C bridge (`i2c_wb_bridge.v`)

Matches the spec:

- `addr` 0-4 map to PRERLO, PRERHI, CTR, TXR/RXR, CR/SR (offsets 0x00-0x10). ✓ (§8.1 asks you to confirm these offsets; they match.)
- `addr` 5-7 return 0xDEADBEEF on read and ignore writes. The bridge hides the core's debug-read registers at 3'b101-111.
- Bits 31:8 read 0.
- Writes without `wstrb[0]` are ignored but still complete.
- `ARST_LVL=1` with `arst_i=~resetsn` gives a correct reset polarity.
- Timing: request at t, ack at t+2, `ready` at t+3. The Wishbone CYC/STB window yields exactly one write per access, and there is no double-fire on the following cycle.
- `irq` is level (`irq_flag && ien`) for `irq[5]`.

Potential concerns

- **I-7:** `scl_drive_low` / `sda_drive_low` need an open-drain IOBUF (or equivalent) at the top level, plus external pull-ups.
- Latency is about 3 cycles, which PicoRV32's `mem_ready` tolerates.

B7. OpenCores core (`i2c_master_*.v`)

Matches the spec:

- PRER reset 0xFFFF.
- SR layout: RXACK[7], BUSY[6], AL[5], TIP[1], IF[0].
- CR layout: STA[7], STO[6], RD[5], WR[4], ACK[3], IACK[0].
- CR[7:4] and IACK self-clear.
- CTR EN[7], IEN[6].

- Module port lists and instantiations across the four files are mutually consistent (static).

Confirmed deviation

- **I-1:** CTR[5:0] is a plain 8-bit register. Spec §8.3 says bits 5:0 are reserved with reads 0 and writes ignored.

Potential concerns

- **I-3: this copy contains slave-mode extensions, not the standard master-only core.**
 - Evidence: `my_addr`, the `sladr` register at adr 3'b111, `slave_en`, `sl_cont`, `slave_dat_req` / `slave_dat_avail`.
 - Impact on the spec register set: CTR[5] becomes `slave_en`, CR[1] becomes `sl_cont`, SR[4:2] become slave status, and IF is also set by slave events.
 - With CTR[5]=0 these should read and behave as reserved, but the spec describes "the standard OpenCores register set". I could not diff against a pristine copy. Recommend doing so, or removing the slave logic if unneeded.
- **I-4:** `assign slave_reset = sta_condition | sto_condition` feeds the byte controller's reset branch, which also clears `master_mode`.
 - The core's own START is seen on the bus, so the byte FSM is reset mid-START. It re-issues START, and the bit controller's completion `cmd_ack` should still advance it.
 - `master_mode` drops for a cycle or two, releasing SCL/SDA briefly. By static trace it works, but a **simulation with a slave model must confirm it**, including repeated-start. No testbench was supplied.
- **I-5:** `i2c_master_defines.v` is pulled in by ``include` in three files. It must be on the include path (Vivado: mark it as a global include or set `include_dirs`).
- **I-6:** the CR write is gated on CTR.EN (standard OpenCores behaviour). The spec does not mention this, and firmware must enable the core first.

C. Missing Blocks and Dependencies

Not supplied (cannot verify):

- Boot ROM (and the bootloader image)
- App RAM (64 KB)
- UART (the bootloader UART)
- Address decoder and IRQ collector (Spec §13 #1 says they must move to the new map)

Also missing:

- Testbenches for every block
- `fir_golden.py`, `dsp.h` (`sw_fir_step`), `coeffs.h`
- A pristine OpenCores copy for diffing
- Any lint, simulation or constraints files

All I2C submodules are present: top, byte_ctrl, bit_ctrl, defines.

D. Interface Compatibility Report

Common slave convention across Timer, GPIO, FIR and I2C: `clk`, `resetn` (active-low, synchronous), `sel`, `wstrb[3:0]`, `wdata[31:0]`, `rdata[31:0]` (registered), `ready` (one-cycle ack), plus an `irq` where applicable. There are no mismatches in widths or polarity.

Block	<code>addr</code> port	Intended source	Notes
Timer	<code>[1:0]</code>	<code>mem_addr[3:2]</code> (not documented in the file)	Aliases above 0x0C
GPIO	<code>[2:0]</code>	<code>mem_addr[4:2]</code> (documented)	
FIR	<code>[4:0]</code>	<code>mem_addr[4:0]</code> , uses <code>[4:2]</code>	Raw byte-address bits, unlike the others
I2C bridge	<code>[2:0]</code>	<code>mem_addr[4:2]</code> (documented)	

The addressing convention is inconsistent across peripherals. It is not a spec violation, but the integrator must slice each one differently.

Other mismatches and notes:

- **IRQ numbering:** Timer `irq[3]`, FIR `irq[4]` and I2C `irq[5]` match Spec §2.2. The architecture diagram labels the MI modem "irq 5", but the spec uses `irq[5]` for I2C and `irq[6]` for the MI modem. Spec §13 #7 already notes the diagram needs an update.
- **Read data:** it is never gated by `sel` in any peripheral, so the integrator must mux by region.
- `sel`: all four rely on `sel` dropping the cycle after `ready`.
- **Decoder:** the diagram says the decoder uses `addr[31:28]`, which gives 256 MB regions, while the spec says 4 KB windows. See A-1.
- **Board-dependent bits:** GPIO needs board tie-offs (G-1).
- **I2C pads:** SCL/SDA need open-drain tri-state at top level (I-7).
- **Timescale:** `timer.v` and `gpio.v` have a ``timescale`; the FIR files do not. This is a simulation warning risk.

E. Final Readiness Summary

1. **Total required hardware blocks:** 10.
2. **Counts:** PASS 3 (GPIO, FIR core, I2C bridge), PARTIAL 3 (Timer, FIR wrapper, OpenCores I2C core), FAIL 0, UNVERIFIED 4 (Boot ROM, App RAM, UART, decoder/IRQ collector).
3. **Blocks needing correction before integration:** Timer, FIR wrapper, FIR core (small fix), I2C core (decision), `fir.h` (firmware).
4. **Exact corrections:**

Block	Correction
Timer	T-1: make the tick-set unconditional so the set wins over a same-cycle clear. T-2: remove or document the dead <code>cnt <= 0</code> in the PERIOD write case.
FIR wrapper	F-1: return 0xDEADBEEF in the <code>default</code> branch (offsets 0x18/0x1C). Consider requiring <code>wstrb == 4'hF</code> for writes (F-3).
FIR core	F-2: reset <code>dout</code> to 0. Fix the "TAPS+4" comment to TAPS+3.
I2C core	I-1: mask CTR to bits [7:6] (read 0 on 5:0) or have the spec accept the extra bits. Decide whether to keep the slave-mode extensions (I-3); if kept, simulate master START/repeated START (I-4).
<code>fir.h</code>	Make <code>fir_init</code> return an error. Make a timeout of 0 non-infinite. Have <code>fir_step</code> return an error indication. Fix the "Rev 0.2" and section references (FH-1 to FH-3).

5. Clarifications needed before compliance can be established:

- **A-1:** Should offsets inside a 4 KB window beyond the listed registers alias (Spec §1) or return 0xDEADBEEF (Spec §1, "unmapped address")? The spec says both. GPIO, FIR and I2C implement DEADBEEF only partway into the window, and all alias after that. Decide on one rule and apply it everywhere.
- **A-2:** Is there a Rev 0.2 of the spec? `fir.h` cites one.
- **A-3:** Is the slave-mode I2C variant intentional? Spec §8 says "standard OpenCores".
- **A-4:** How many buttons exist on Boolean: 4 (spec "confirm") or 5? The RTL is 5-bit.
- **A-5:** Where do the UART, Boot ROM, App RAM and decoder RTL live? Please supply them for audit.
- **A-6:** Supply the testbenches and golden model (`sw_fir_step` , coefficients) so the FIR "SIM-VERIFIED" and Timer/I2C claims can be checked rather than assumed.